

ACP

AARAMBH CHAPTERWISE PROBLEMS

Class-9th

DO NOT CROSS POLICE LINE

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Force and Laws of Motion

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FORCE AND LAWS OF MOTION

 (Options acche se padhna)

MULTIPLE CHOICE QUESTIONS (1 Mark)

1. A body of mass 5 kg is acted upon by a force of 20 N. What is the acceleration produced in the body?
 - (a) 2 m/s^2
 - (b) 4 m/s^2
 - (c) 5 m/s^2
 - (d) 20 m/s^2
2. A cricket player lowers his hands while catching the ball. This is done to:
 - (a) increase the force
 - (b) reduce momentum
 - (c) increase time of contact
 - (d) reduce mass of ball
3. A car is moving with uniform velocity. Which of the following statements is true?
 - (a) No force is acting on the car
 - (b) Balanced forces are acting on the car
 - (c) Unbalanced force is acting
 - (d) Friction is increasing the velocity
4. What would happen if the smaller ball were rolling with a velocity of 5 m/s and struck the bigger ball at rest?
 - (a) The two balls would continue to roll in the direction of the strike
 - (b) The smaller ball would rebound and the bigger ball would roll forward
 - (c) The two balls would roll in the direction opposite to the strike
 - (d) The smaller ball would stop rolling and the bigger ball would start rolling.
5. An object of mass 2 kg is sliding with a constant velocity of 4 m/s on a frictionless horizontal table. The force required to keep the thing moving at the same speed is

- (a) 2 N
- (b) 0 N
- (c) 32 N
- (d) 8 N

6. Which of the following has the greatest inertia?

- (a) A rubber ball
- (b) A 5 kg iron block
- (c) A tennis ball
- (d) A wooden stick

7. A body is initially at rest. It is acted upon by a constant force. The work done by the force depends on:

- (a) Only displacement
- (b) Only mass
- (c) Displacement and force
- (d) Acceleration

8. A man jumps out of a moving train. His feet hurt due to:

- (a) Reduced weight
- (b) Sudden change in inertia
- (c) Reaction force
- (d) Increase in velocity

9. The momentum of a body is doubled by:

- (a) Doubling its mass
- (b) Doubling its velocity
- (c) Doubling both mass and velocity
- (d) Either (a) or (b)

10. Newton's First Law is also known as:

- (a) Law of Acceleration
- (b) Law of Inertia
- (c) Law of Momentum
- (d) Action-Reaction Law

11. A force of 100 N is applied on a body of mass 20 kg. What is the acceleration?

- (a) 2 m/s^2
- (b) 5 m/s^2
- (c) 10 m/s^2

(d) 20 m/s^2

12. A water tanker filled up to $\frac{2}{3}$ of its height is moving with uniform speed. On sudden application of brakes, the water in the tank would:

- (a) move backward
- (b) move forward
- (c) be unaffected
- (d) rise upwards

13. Angelina throws a stone of 2 kg on the road. The stone with a velocity of 20 m/s. The stone travels a distance of 40 m. What is the force opposing the motion of the stone on the road?

- (a) Friction force, 10 N
- (b) Gravitational force, 10 N
- (c) Inertia of rest, 10 N
- (d) Frictional force, -10 N

14. Inertia is the property of a body to:

- (a) Change its motion
- (b) Remain at rest only
- (c) Remain in uniform motion or rest unless acted upon by an external force
- (d) Resist gravity

15. If force is halved and mass is doubled, what happens to acceleration?

- (a) Doubled
- (b) Halved
- (c) Reduced to one-fourth
- (d) Remains same

16. Why do seat belts reduce the risk of injury in a car accident?

- (a) Increases momentum
- (b) Increases friction
- (c) Applies backward force
- (d) Increases time of impact, reducing force

17. Two people push a box from opposite sides with equal force. The box does not move. What is the net force?

- (a) Zero
- (b) Equal to one of the forces
- (c) Sum of forces
- (d) Depends on friction

18. An object of mass 2 kg is sliding with a constant velocity of 4 m/s on a frictionless horizontal table. The force required to keep the object moving with the same velocity is

- (a) 32 N
- (b) 0 N
- (c) 12 N
- (d) 8 N

 (pahle points socho firr likho)

VERY SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)

19. We want to move a wooden cabinet across the floor with a horizontal force of 200N at a constant speed. What is the frictional force exerted on the case?

20. There are three bodies made of aluminium, steel and wood that have the same shape and volume. Which of them would have greater inertia?

21. Two balls of the same size but different materials, rubber and iron, are held on the smooth floor of a moving train. The brakes are suddenly applied to stop the train. Are the balls rolling? If yes, in which direction? Will they move at the same speed? Justify your answer.

22. Two identical bullets are fired, one from a light rifle and another one from a heavy rifle of the same power. Which rifle will hurt the shoulder more, and why?

23. A horse continues to exert force to move a cart at a constant speed. Explain why.

24. Suppose a ball of the mass m is thrown vertically upwards with an initial velocity v , where its velocity decreases continuously until it becomes zero. After that, the ball starts falling and reaches the speed v before hitting the ground. That implies that the magnitude of the ball's initial and final moments are equal. However, it is not an example of conservation of momentum. Explain why.

25. Two friends on roller skates face each other 5 m apart. One throws a 2kg ball to the other, who catches it. How does this activity affect the position of the two? Explain your answer

26. The lawn sprinkler starts rotating as soon as the water is added. Explain the principle on which it works.

(pahle points socho firr likho)

SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)

27. Which has more inertia:

- (a) a rubber ball and a rock of the same size?
- (b) a bicycle and a train?
- (c) a five rupee coin and a one rupee coin?

28. Using a horizontal force of 200 N, a man intends to move a wooden cabinet across a floor at a constant velocity. What is the frictional force that will be exerted on the cabinet?



29. Explain why some leaves may fall from a tree when its branch is violently shaken.

30. Why do you fall forward when a moving bus slows down and fall back when it accelerates from a standstill

31. Explain how a horse can pull a cart if action is always equal to the reaction.

32. Explain why it is difficult for a firefighter to hold onto a hose discharging large volumes of water at high speed.

33. A bullet with a mass of 50 g is fired from a gun with a mass of 4 kg with an initial velocity of 35 m s^{-1} . Calculate the rifle's initial recoil velocity.

34. Two trucks of mass equal to 5000 kg are moving on a road. One is moving with a velocity of 50 m/s while the other is moving with a velocity of 2 m/s. Calculate the force required to stop them in 20 seconds. Which one of them requires more force and why?

35. During the school trip, the bus was covering some distance in a given time, which is recorded by Ram as per the assignment given by his teacher. The following is distance-time table of the bus when in motion:

Time	Distance
0	0
1	1
2	8
3	27
4	64

36. During an army training session, each soldier fires 'n' bullets per second from a machine gun aimed at a target. The mass of each bullet is 'm', and the velocity at which they are shot is 'v' m/s. What force must the soldier apply to keep the gun steady in position to successfully hit the target?

37. In the school athletic meet, Gargi saw that before the start of the race, runners press the ground with their feet. What is the reason behind it?

38. Determine the time required by an object having weight 10 kg to gain the velocity of 80 m/s by applying force of 50 N.

39. Which of the following has more inertia?

- (A) A rubber ball and a stone of the same size.
- (B) A bicycle and a train.
- (C) A five rupees coin and a one-rupee coin.

40. In a fair, a child buys a balloon, but he unties the balloon. What will be observed when the balloon is untied?

41. Why Newton's first law of motion is known as the Law of Inertia?

42. An object of mass 50 kg accelerates at 3 m/s^2 . Calculate the force applied to the object.

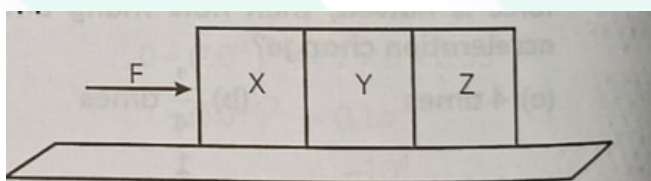
(A) A man throws a ball of 0.4 kg vertically upward with a velocity of 15 m/s. What will be its momentum at maximum height? Define Newton's first law of motion

(pahle points socho firr likho)

LONG ANSWER TYPE QUESTIONS (5 MARKS)

43. Seema pushed three identical blocks, each having a mass M on a frictionless table and applied the force as F .

- (A) What is the acceleration of the blocks?
- (B) What is the net force on block 'X'?
- (C) What force does 'X' apply on 'Y'?
- (D) What force does 'Y' apply on 'Z'?
- (E) Show action-reaction pairs on the contact surfaces of the blocks?



44. An 8000 kg locomotive pulls a train of 5 cars on a horizontal track, each weighing 2000 kg. If the engine exerts a force of 40000N and the trail provides a frictional force of 5000N, then calculate

- (a) The net acceleration force
- (b) The acceleration of the train

45. A hockey ball with a mass of 200 g and moving at 10 ms^{-1} is hit by a hockey stick to return it to its original path at a speed of 5 ms^{-1} . Calculate the change in momentum as the hockey ball moves due to the force exerted by the hockey stick.

46. (a) State the reason why a bullet of small mass fired from a gun kills a person.

(b) A bullet of mass 4 g fired with a velocity of $50 \text{ m}\cdot\text{s}^{-1}$ enters a wall up to a depth of 10 cm. Calculate the average resistance offered by the wall.

(c) How will the depth of penetration into the wall change if a bullet of mass 5 g strikes against it with a velocity of $40 \text{ m}\cdot\text{s}^{-1}$? Give reason to justify your answer.

47. Two objects A and B have masses 100 g and 200 g moving in the same direction with speeds 2 m/s and 1 m/s respectively. They collide. After collision, A moves with a speed of 1.67 m/s. Find the velocity of B after collision. State which law of motion is used.

48. A 1 kg stone is thrown over the frozen surface of a lake at a speed of 20 ms^{-1} and stops after a distance of 50 m. What is the frictional force between the stone and the ice?
49. An object with a mass of 100 kg accelerates smoothly from a speed of 5 ms^{-1} to 8 ms^{-1} in 6s. Calculate the initial and final momentum of the object. Also, find the magnitude of the force being exerted on the object.
50. Derive the law of conservation of momentum from Newton's third law.
51. Why is it dangerous to jump out of a moving bus? Why does a runner continue to run forward even after crossing the finish line? Explain using laws of motion.

ASSERTION AND REASON TYPE QUESTIONS

- (A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
- (B) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.
- (C) Assertion is true, but the Reason is false.
- (D) Assertion is false, but the Reason is true.

52. Assertion (A): Water sprinkler used for grass lawns begins to rotate as soon as the water is supplied

Reason (R): As soon as water comes out of the nozzle of the sprinkler, it exerts an equal force on the nozzle in opposite direction.

53. Assertion (A): When a car rounds a curve suddenly, a person sitting inside it is thrown outwards.

Reason (R): When a car turns, the person tries to maintain his original direction of motion due to inertia of rest.

54. Assertion (A): A heavier object requires more force to move than a lighter one.

Reason (R): According to Newton's Second Law, acceleration is directly proportional to mass.

CASE STUDY BASED QUESTION

55. The behaviour of a heavy body at rest or in uniform linear motion, i.e., not accelerating or revolving, is described by Isaac Newton's First Law of Motion. "A body at rest will remain at rest, and a body in motion will remain in motion unless it is acted upon by an external force." This simply means that things cannot start, stop, or change direction on their own. To bring about such a change, they must be subjected to some external force. While this concept may appear easy and clear to us today, it was groundbreaking in Newton's time. In the field of aerodynamics, there are numerous outstanding illustrations of Newton's first law. The first law describes the motion of an aeroplane when the pilot changes the throttle setting of the engine. Newton's first law can be seen in the motion of a ball falling through the atmosphere or a model rocket being launched into the atmosphere. The first law can also be used to describe the motion of a kite as the wind changes. Thus, Newton's First Law is not just a theoretical concept but is observed in various real-life scenarios, from sports to space travel.

A) When a balloon is pressed between palms, the shape changes. It happens due to:

- (a) gravitational force
- (b) unbalanced force
- (c) balanced force
- (d) frictional force

B) Why does a horseback rider fall when the horse suddenly starts running?

- (a) The rider has developed a sudden fear of falling.
- (b) The rider's lower body travels forward with the horse, but the upper body remains still due to inertia.
- (c) The rider is pushed to the right.
- (d) None of the above

C) Assertion (A): A body is moving along a straight line with constant speed and the force is applied to the moving body.

Reason (R): The body will continue to move in the same direction.

- A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
- (B) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.
- (C) Assertion is true, but the Reason is false.
- (D) Assertion is false, but the Reason is true.

PK SPECIAL [KBC]



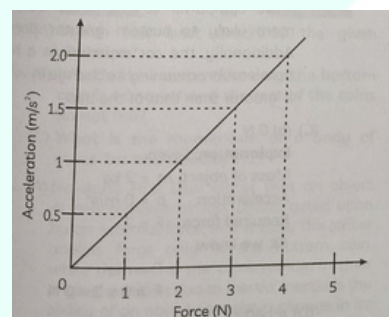
56. Yogesh measures the acceleration of the Volvo using an apparatus. The result is obtained as shown in the Acceleration–Force graph. It is known that the more is force applied to an object, the more is its acceleration.

Acceleration is proportional to force, as may be demonstrated experimentally. It may also be demonstrated that the greater the object's mass, the lower is its acceleration for the same force. The force must rise in proportion to the mass to obtain the same acceleration.

(A) Calculate the mass of Volvo from given graph if Yogesh takes the force applied as 1 N.

(B) If the force applied on the Volvo is doubled and acceleration is halved, what is the ratio of mass?

(C) define momentum . what happens to the momentum of a body if speed is doubled?



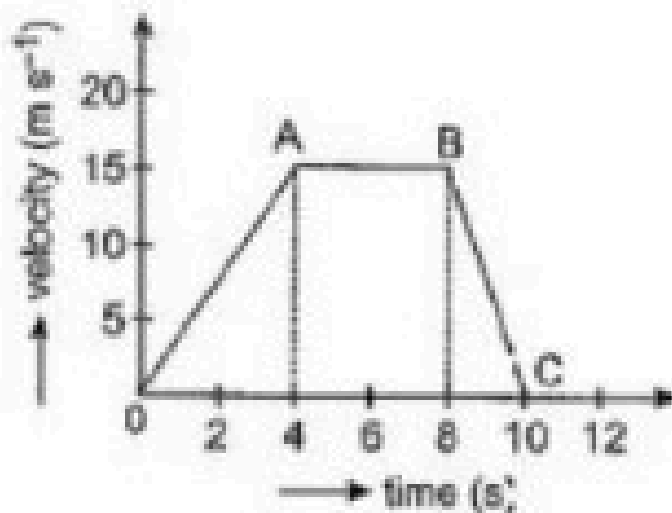
57. Two bodies of masses 2 kg and 4 kg are moving towards each other with velocities 3 m/s and 2 m/s, respectively. They collide and stick together. Find their common velocity after collision.

58. A person standing on a cart throws a ball forward. What happens to the cart and why? Assume no external force (like friction) acts.

59. A cricket ball of mass 150 g moving at 20 m/s is hit by a bat and returns along the same line with a velocity of 30 m/s. If the time of contact between bat and ball is 0.01 s, calculate the force exerted by the bat.

60. The velocity-time graph of a car of 1000 kg mass is given alongside. From the graph answer the following: [2024]

- (a) When is the maximum force acting on the car? Why?
- (b) What is the retarding force?
- (c) For how long is there no force acting?



Kitne Questions kar paaye?

SOLUTIONS



SOLUTIONS

MULTIPLE CHOICE TYPE QUESTIONS:

1. (b)
2. (c)
3. (b)
4. (b)
5. (b)
6. (b)
7. (c)
8. (b)
9. (d)
10. (b)
11. (b)
12. (b)
13. (d)
14. (c)
15. (c)
16. (d)
17. (a)
18. (b)

VERY SHORT ANSWER TYPE QUESTIONS:

19. The frictional force is equal and opposite to the applied force, i.e., 200 N (opposite to motion direction).

20. Steel has greater inertia (indolence) because it has the highest mass among the three.

21. Yes, both balls roll forward due to inertia.

Iron ball moves faster as it has more mass and hence more inertia.

22. The light rifle will hurt more because it has less mass, causing more recoil (backward acceleration).

23. The horse continues to apply force to balance the friction and keep the cart moving at constant speed.

24. It's not an example of momentum conservation because external force (gravity) acts on the ball during motion.

25. Both friends will move apart slightly.

This happens due to the law of conservation of momentum – when the first friend throws the ball, they move backward, and when the second catches it, they also move backward.

26. The lawn sprinkler works on Newton's Third Law of Motion –

For every action (water rushing out), there is an equal and opposite reaction (sprinkler rotates).

SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)

27.(a) The rock has more inertia than a rubber ball of the same size because it has more mass.

(b) The train has more inertia than a bicycle due to its much larger mass.

(c) The five-rupee coin has more inertia than the one-rupee coin, as inertia is directly proportional to mass.

28. When the cabinet is moving at a constant velocity, the net force acting on it is zero.

Hence, the frictional force opposing the motion is equal and opposite to the applied force.

So, the frictional force exerted on the cabinet is 200 N.

29. When the branch of a tree is shaken suddenly, the branch moves, but the leaves tend to remain at rest due to the inertia of rest.

As a result, the connection between some leaves and the branch breaks, causing the leaves to fall.

30. When a moving bus suddenly slows down, our body tends to keep moving forward due to inertia, making us fall forward.

When the bus starts moving suddenly, our body tends to stay at rest and we fall backward.

This happens due to inertia of motion and rest, respectively.

31. According to Newton's Third Law of Motion, when the horse pushes the ground backward with its feet, the ground pushes the horse forward with an equal and opposite force.

This reaction helps the horse to move forward and pull the cart.

32. When a hose ejects water at high speed, it exerts an equal and opposite reaction force on the firefighter holding it.

This makes it difficult to control the hose. The greater the speed and volume of water, the greater the backward force experienced.

33. Mass of bullet (m_1) = 0.05 kg

Velocity of bullet (v_1) = 35 m/s

Mass of gun (m_2) = 4 kg

Let recoil velocity of gun = v_2

Using conservation of momentum:

$$m_1 \times v_1 = m_2 \times v_2$$

$$0.05 \times 35 = 4 \times v_2$$

$$1.75 = 4v_2$$

$$v_2 = 1.75 \div 4 = 0.4375 \text{ m/s}$$

So, the recoil velocity of the rifle is 0.4375 m/s in the opposite direction.

34. Given:

Mass of each truck (m) = 5000 kg

Time (t) = 20 s

For Truck 1:

Initial velocity (u_1) = 50 m/s

Force:

$$F_1 = (m \times u_1) / t$$

$$F_1 = (5000 \times 50) / 20 = 12500 \text{ N}$$

For Truck 2:

Initial velocity (u_2) = 2 m/s

Force:

$$F_2 = (m \times u_2) / t$$

$$F_2 = (5000 \times 2) / 20 = 500 \text{ N}$$

Conclusion:

Truck 1 requires more force because it has a higher velocity and therefore greater momentum, which needs more force to reduce to zero in the same time.

35. We observe that the distance covered follows the pattern:

$$\text{Distance} = \text{Time}^3$$

That is:

$$\text{At } t = 1 \text{ s} \rightarrow s = 1^3 = 1 \text{ m}$$

$$\text{At } t = 2 \text{ s} \rightarrow s = 2^3 = 8 \text{ m}$$

$$\text{At } t = 3 \text{ s} \rightarrow s = 3^3 = 27 \text{ m}$$

$$\text{At } t = 4 \text{ s} \rightarrow s = 4^3 = 64 \text{ m}$$

Since the distance-time graph is not a straight line, the motion is non-uniform. The bus is undergoing accelerated motion because the distance increases more rapidly with time.

36. According to Newton's Second Law of Motion,
Force (F) = Rate of change of momentum.

Each bullet has mass = m

Velocity = v

Number of bullets fired per second = n

Momentum of one bullet = $m \times v$

Total momentum per second = $n \times m \times v$

$\therefore$ Force required to keep the gun steady:

$$F = n \times m \times v$$

The soldier must apply a force of $n \times m \times v$ in the opposite direction to balance the backward recoil and keep the gun steady.

37. According to Newton's Third Law, runners press the ground backward to get an equal and opposite reaction force from the ground.

This reaction helps them to move forward quickly at the start of the race.

38. Given:

Mass (m) = 10 kg

Force (F) = 50 N

Final velocity (v) = 80 m/s

Initial velocity (u) = 0 m/s (since object starts from rest)

Step 1: Calculate acceleration using Newton's Second Law

$$F = m \times a$$

$$\Rightarrow a = F / m = 50 / 10 = 5 \text{ m/s}^2$$

Step 2: Use the first equation of motion

$$v = u + a \times t$$

$$\Rightarrow 80 = 0 + 5 \times t$$

$$\Rightarrow t = 80 / 5 = 16 \text{ seconds}$$

39.(A) A stone has more inertia than a rubber ball of the same size because it has more mass.

(B) A train has more inertia than a bicycle as it has much greater mass.

(C) A five-rupees coin has more inertia than a one-rupee coin because it is heavier.

40. When the balloon is untied, air rushes out in one direction.
Due to Newton's Third Law, the balloon moves in the opposite direction.

41. Newton's First Law states that an object remains at rest or in uniform motion unless acted upon by an external force.
This law explains the inertia of objects, so it is also called the Law of Inertia.

42. Part 1: Force Calculation

Given:

Mass (m) = 50 kg

Acceleration (a) = 3 m/s²

Using Newton's Second Law of Motion:

$$F = m \times a = 50 \times 3 = 150 \text{ N}$$

So, the force applied on the object is 150 newton.

Part 2: Momentum at Maximum Height

Mass of ball = 0.4 kg

Velocity at maximum height = 0 m/s (since the ball stops momentarily before falling back)

$$\text{Momentum} = \text{mass} \times \text{velocity} = 0.4 \times 0 = 0 \text{ kg}\cdot\text{m/s}$$

So, the momentum of the ball at maximum height is 0 kg·m/s.

LONG ANSWER TYPE QUESTIONS (5 MARKS)

43. (A) What is the acceleration of the blocks?

$$\text{Total mass} = M + M + M = 3M$$

$$\text{Applied force} = F$$

Using Newton's Second Law:

$$F = ma$$

$$\Rightarrow a = F / 3M$$

$$\therefore \text{Acceleration, } a = F / 3M$$

(B) Block X pushes both Y and Z (mass = 2M)

So,

$$F_x = 2M \times a$$

$$= 2M \times (F / 3M)$$

$$= 2F / 3$$

$$\therefore \text{Net force on block X} = 2F / 3$$

To find the force X applies on Y, consider the two blocks Y + Z as a system.

The acceleration of the system is:

$$a = F/3M$$

Combined mass of Y and Z = 2M

Force by X on Y = $2M \times a$

$$= 2M \times (F/3M)$$

$$= (2F)/3$$

(D) Mass of Z = M

$$a = F / 3M$$

$$F_{YZ} = M \times a$$

$$= M \times (F / 3M)$$

$$= F / 3$$

$$\therefore \text{Force by Y on Z} = F / 3$$

(E) Action-Reaction Pairs on Contact Surfaces:

Between X and Y:

- X applies $2F / 3$ on Y $\rightarrow$ (Action)
- Y applies $2F / 3$ on X $\leftarrow$ (Reaction)

Between Y and Z:

- Y applies $F / 3$ on Z $\rightarrow$ (Action)
- Z applies $F / 3$ on Y $\leftarrow$ (Reaction)

These are equal and opposite forces, as stated in Newton's Third Law of Motion.

44. Given:

Mass of engine (locomotive),

$$m_e = 8000 \text{ kg}$$

Mass of each car = 2000 kg

Number of cars = 5

Total mass of train:

$$m = m_e + 5 \times 2000$$

$$m = 8000 + 10000 = 18000 \text{ kg}$$

Force exerted by the engine:

$$F = 40000 \text{ N}$$

Frictional force:

$$f = 5000 \text{ N}$$

Total mass of train:

$$m = m_e + 5 \times 2000$$

$$m = 8000 + 10000 = 18000 \text{ kg}$$

Force exerted by the engine:

$$F = 40000 \text{ N}$$

Frictional force:

$$f = 5000 \text{ N}$$

(a) Net Accelerating Force

We know:

$$F_{\text{net}} = F - f$$

$$F_{\text{net}} = 40000 \text{ N} - 5000 \text{ N}$$

$$F_{\text{net}} = 35000 \text{ N}$$

$$\therefore \text{Net accelerating force} = 35000 \text{ N}$$

(b) Acceleration of the Train

Using Newton's Second Law:

$$F = m \times a$$

$$\Rightarrow a = F_{\text{net}} / m$$

$$a = 35000 / 18000$$

$$a = 1.94 \text{ m/s}^2$$

$$\therefore \text{Acceleration of the train} = 1.94 \text{ m/s}^2$$

45. Given:

$$m = 0.2 \text{ kg}$$

$$u = +10 \text{ m/s}$$

$$v = -5 \text{ m/s}$$

Formula:

$$\Delta p = m \times (v - u)$$

$$\Delta p = 0.2 \times (-5 - 10)$$

$$\Delta p = 0.2 \times (-15)$$

$$\Delta p = -3 \text{ kg}\cdot\text{m/s}$$

Answer:

$$\text{Change in momentum} = -3 \text{ kg}\cdot\text{m/s}$$

$$\text{Magnitude of change in momentum} = 3 \text{ kg}\cdot\text{m/s}$$

46.(a) A bullet, though small in mass, has a very high velocity. Due to this, it possesses a large momentum. When it strikes a person, it transfers this large momentum in a short time, resulting in a very large force that can kill the person.

46.(a) A bullet, though small in mass, has a very high velocity. Due to this, it possesses a large momentum. When it strikes a person, it transfers this large momentum in a short time, resulting in a very large force that can kill the person.

(b) Given:

$$\text{Mass (m)} = 4 \text{ g} = 0.004 \text{ kg}$$

$$\text{Velocity (u)} = 50 \text{ m/s}$$

$$\text{Penetration depth (s)} = 10 \text{ cm} = 0.1 \text{ m}$$

$$\text{Final velocity (v)} = 0 \text{ (as it comes to rest)}$$

Using the equation:

$$v^2 = u^2 + 2as$$

$$0 = 50^2 + 2 \times a \times 0.1$$

$$\Rightarrow a = -12500 \text{ m/s}^2$$

Now,

$$F = ma = 0.004 \times (-12500) = -50 \text{ N}$$

Answer: The average resistance offered by the wall is 50 N.

(c) New mass = 5 g = 0.005 kg

$$\text{New velocity} = 40 \text{ m/s}$$

$$\text{Kinetic Energy} = (1/2)mv^2$$

$$= (1/2) \times 0.005 \times (40)^2 = 4 \text{ J}$$

The initial bullet had more energy (5 J), so the second bullet with 4 J energy will penetrate less.

Answer: The bullet will penetrate less than 10 cm due to its lower kinetic energy.

47. Given:

$$\text{Mass of A} = 100 \text{ g} = 0.1 \text{ kg}$$

$$\text{Initial velocity of A, } u_1 = 2 \text{ m/s}$$

$$\text{Mass of B} = 200 \text{ g} = 0.2 \text{ kg}$$

$$\text{Initial velocity of B, } u_2 = 1 \text{ m/s}$$

$$\text{Final velocity of A, } v_1 = 1.67 \text{ m/s}$$

$$\text{Let final velocity of B} = v_2$$

Using conservation of momentum:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$(0.1 \times 2) + (0.2 \times 1) = (0.1 \times 1.67) + (0.2 \times v_2)$$

$$0.2 + 0.2 = 0.167 + 0.2v_2$$

$$0.4 - 0.167 = 0.2v_2$$

$$0.233 = 0.2v_2$$

$$v_2 = 0.233 / 0.2 = 1.165 \text{ m/s}$$

48. Mass of the stone (m) = 1 kg

Initial velocity (u) = 20 m/s

Final velocity (v) = 0 (since it stops)

Distance (s) = 50 m

Using the equation of motion:

$$v^2 = u^2 + 2as$$

$$0 = (20)^2 + 2 \times a \times 50$$

$$0 = 400 + 100a$$

$$a = -4 \text{ m/s}^2$$

Frictional force (F) = m × a = 1 × (-4) = -4 N

Answer: The frictional force between the stone and the ice is 4 N (opposing motion).

49. Mass (m) = 100 kg

Initial speed (u) = 5 m/s

Final speed (v) = 8 m/s

Time (t) = 6 s

Initial momentum = m × u = 100 × 5 = 500 kg·m/s

Final momentum = m × v = 100 × 8 = 800 kg·m/s

Change in momentum = 800 – 500 = 300 kg·m/s

Force = Change in momentum / Time = 300 / 6 = 50 N

Answer:

Initial momentum = 500 kg·m/s

Final momentum = 800 kg·m/s

Force applied = 50 N

50. Let two objects A and B have masses m_1 and m_2 , and initial velocities u_1 and u_2 respectively.

Let their final velocities after collision be v_1 and v_2 .

According to Newton's third law:

Force on A by B = – Force on B by A

Using Newton's second law,

Force on A = $m_1(v_1 - u_1)/t$

Force on B = $m_2(v_2 - u_2)/t$

So,

$$m_1(v_1 - u_1)/t = -m_2(v_2 - u_2)/t$$

$$\Rightarrow m_1(v_1 - u_1) = -m_2(v_2 - u_2)$$

$$\Rightarrow m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Total momentum before collision = Total momentum after collision

This is the Law of Conservation of Momentum, derived using Newton's third law.

51. When a person jumps out of a moving bus, the lower part of the body comes to rest on touching the ground, but the upper part continues to move due to inertia of motion. This can cause the person to fall forward and get hurt. Similarly, a runner continues to run forward even after crossing the finish line due to the same reason. His body remains in motion due to inertia and needs time or external force (like friction) to stop.

Answer:

This behavior is explained by Newton's First Law of Motion, which states that a body continues in its state of rest or uniform motion unless acted upon by an external force.

ASSERTION AND REASON TYPE QUESTIONS

52. (A)

53. (A)

54. (C)

CASE STUDY BASED QUESTION

55. A) (b) unbalanced force

B) (b) The rider's lower body travels forward with the horse, but the upper body remains still due to inertia.

C) (C) Assertion is true, but the Reason is false.

PK SPECIAL [KBC]

56. (A) From Newton's Second Law of Motion:

$$F = ma$$

$$\Rightarrow m = F / a$$

Given:

Force, $F = 1 \text{ N}$

Acceleration = a (from graph)

Then,

Mass of Volvo, $m = 1 / a \text{ kg}$

(B) Let initial force = F , acceleration = a , mass = m_1

Then:

$$F = m_1 \times a \Rightarrow m_1 = F / a$$

Now, new force = $2F$, new acceleration = $a / 2$

Let new mass = m_2

Using formula:

$$2F = m_2 \times (a / 2)$$

$$\Rightarrow m_2 = (2F \times 2) / a = 4F / a$$

Now,

$$m_1 = F / a$$

So,

$$m_2 : m_1 = (4F / a) : (F / a) = 4 : 1$$

Therefore, the ratio of masses is $4 : 1$

(C) Momentum is defined as the product of mass and velocity of an object.

$$\text{Momentum (p)} = m \times v$$

If speed is doubled,

$$\text{New momentum} = m \times 2v = 2mv$$

So, momentum also gets doubled.

57. Given:

Mass of first body, $m_1 = 2 \text{ kg}$

Velocity of first body, $u_1 = +3 \text{ m/s}$

Mass of second body, $m_2 = 4 \text{ kg}$

Velocity of second body, $u_2 = -2 \text{ m/s}$ (opposite direction)

After collision, they stick together.

Let common velocity = v

Using law of conservation of momentum:

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) \times v$$

$$(2 \times 3) + (4 \times -2) = (2 + 4) \times v$$

$$6 - 8 = 6v$$

$$-2 = 6v$$

$$v = -2 / 6 = -1/3 \text{ m/s}$$

The common velocity after collision is $-1/3 \text{ m/s}$.

58. A person standing on a cart throws a ball forward.

According to the law of conservation of momentum, when the person throws a ball forward, the cart moves backward. This is because initially the total momentum was zero. To conserve momentum, the backward

58. A person standing on a cart throws a ball forward.

According to the law of conservation of momentum, when the person throws a ball forward, the cart moves backward. This is because initially the total momentum was zero. To conserve momentum, the backward momentum of the cart and person balances the forward momentum of the ball.

59. A cricket ball of mass 150 g moving at 20 m/s is hit by a bat and returns along the same line with a velocity of 30 m/s. Time of contact = 0.01 s. Calculate the force exerted by the bat.

Given:

Mass of ball, $m = 150 \text{ g} = 0.15 \text{ kg}$

Initial velocity, $u = 20 \text{ m/s}$

Final velocity, $v = -30 \text{ m/s}$ (opposite direction)

Time, $t = 0.01 \text{ s}$

Change in momentum = $m(v - u)$

$$= 0.15 \times (-30 - 20)$$

$$= 0.15 \times (-50) = -7.5 \text{ kg}\cdot\text{m/s}$$

Force = Change in momentum / Time

$$F = -7.5 / 0.01 = -750 \text{ N}$$

The force exerted by the bat is 750 N in the opposite direction.

60. (a) In the v-t graph, the OA part represents uniformly accelerated motion with an acceleration:

$$a = 30 / 8 = 3.75 \text{ m}\cdot\text{s}^{-2}$$

Force in this region is maximum, having a value:

$$F = m \times a = 1000 \times 3.75 = 3750 \text{ N}$$

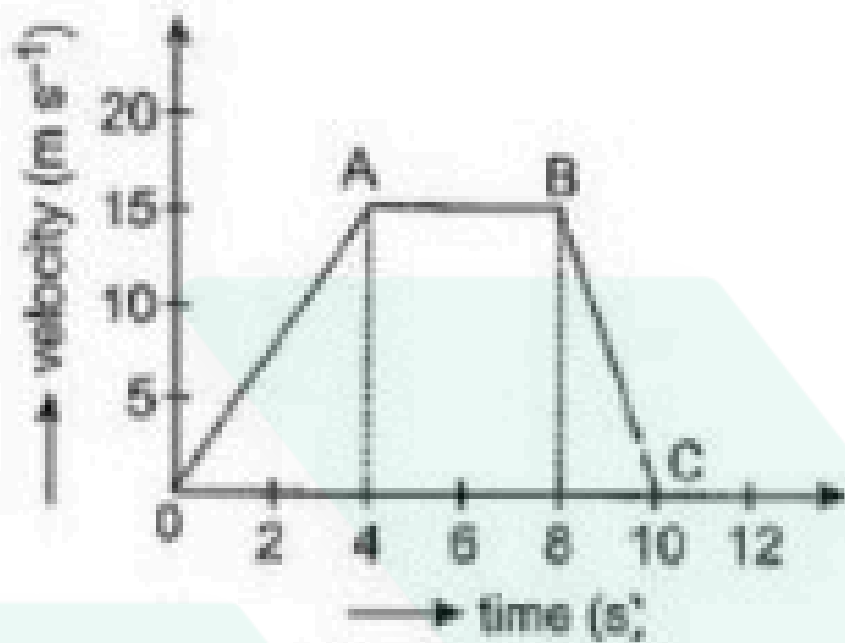
(b) In the v-t graph, the part BC represents uniformly retarded motion with an acceleration:

$$a = -30 / 4 = -7.5 \text{ m}\cdot\text{s}^{-2}$$

Retarding force:

$$F = m \times a = 1000 \times (-7.5) = -7500 \text{ N}$$

(c) The region AB of the graph represents uniform motion with a constant velocity of $15 \text{ m}\cdot\text{s}^{-1}$. Hence, there is no force acting on the car during the part AB from 4 s to 8 s.

**Fig. 9.9**